An embedded eigenvalue in an asymptotically flat purely twisted strip

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Abstract

We give a counterexample to the universal absence assertion in AIM problem 1007. An explicit one-parameter family of positive twisting profiles contains an embedded eigenvalue for every sufficiently large value of a scale parameter. The proof treats the complete transverse operator, uses a contraction in global Sobolev spaces, and enforces the longitudinal matching condition through a continuous antiperiodic phase family.

Target and status. This is a proposed negative resolution of AIM problem 1007, at repository revision `ee5c5c52f527`. It concerns the point-spectrum question strictly above the first transverse threshold; the separate resonance question in the source is outside this target. The accompanying audit records a fresh AI review and symbolic checks, not a human peer review or a formal verification.

For $a > 0$ let $\Omega_a = \mathbb{R} \times (-a, a)$ and $E_j = (j\pi/(2a))^2$. The admissible class consists of $q \in L^\infty(\mathbb{R})$ such that $q \geq 0$, q is not almost everywhere zero, $a\,\|q\|_\infty \leq \sqrt{2}$, and $\operatorname{ess\,sup}_{|s|>R} q(s) \to 0$ as $R \to \infty$. With $f_q(s,t) = \sqrt{1 + t^2 q(s)^2}$, let H_q be the selfadjoint operator in $L^2(\Omega_a, f_q\,ds\,dt)$ defined by

$$h_q[\psi] = \int_{\Omega_a} (f_q^{-1}|\psi_s|^2 + f_q|\psi_t|^2)\,ds\,dt, \qquad D(h_q) = H_0^1(\Omega_a).$$

We denote this admissible class by $\mathcal{Q}_a$.

Theorem 1. *Set*

$$m = \frac{1}{3} - \frac{2}{\pi^2}, \qquad k = \frac{1}{\sqrt{2m}}, \qquad \lambda_* = \frac{\pi^2}{4} + k^2 = \frac{\pi^4}{4(\pi^2 - 6)}.$$

There is a finite S_0 such that, for every $S \geq S_0$, some $\delta \in [0, 2\pi]$ has the following property. The profile

$$q_{S,\delta}(s) = \left[\frac{16k}{S + |s|}\left(1 + \frac{1}{2}\cos(2k|s| + \delta)\right)\right]^{1/2} \tag{1}$$

belongs to $\mathcal{Q}_1$, and $H_{q_{S,\delta}}$ has a nonzero eigenfunction in $H^2(\mathbb{R} \times (-1,1)) \cap H_0^1(\mathbb{R} \times (-1,1))$ at energy λ_*. In particular, $E_1 < \lambda_* < E_2$. For every $a > 0$, the profile $q_a(s) = a^{-1} q_{S,\delta}(s/a)$ belongs to $\mathcal{Q}_a$ and has an embedded eigenvalue $a^{-2}\lambda_*$.

Only the phase is selected by existence. It is specified below as a zero of a continuous real function obtained from convergent linear inverses. No transverse truncation or numerical spectral assertion is used.

The intrinsic Dirichlet Laplace–Beltrami operator describes a particle on a two-dimensional twisted ribbon, confined transversely by its edges but free to propagate longitudinally. This is a surface or ribbon quantum waveguide; the result concerns this model rather than a solid three-dimensional twisted tube. In the asymptotic free guide, the transverse mode χ_j has

dispersion $\lambda = E_j + p^2$, where p is longitudinal momentum. A channel is open when $\lambda > E_j$, permitting longitudinal propagation. Since $\lambda_* \in (E_1, E_2)$, the first channel is open, and the constructed state's component $u\chi_1$ in that channel is nonzero. Nevertheless, its eigenfunction is square integrable in $f_q\,ds\,dt$ and defines a normalizable stationary state: a bound state in the continuum. Localization concerns the probability distribution, not compact support. Under Schrödinger evolution, the eigenstate changes only by a global phase, so its probability density remains stationary despite the available longitudinal escape channel. The construction achieves this through oscillatory long-range twisting.

1 Half-strip formulation and uniform estimates

Work with $a = 1$. For $J = (-1, 1)$ define the transverse Hilbert space and Dirichlet operator by

$$\mathfrak{h} = L^2(J, dt), \qquad T = -\partial_t^2, \qquad D(T) = H^2(J) \cap H_0^1(J).$$

Its normalized orthonormal eigenbasis $(\chi_j)_{j\geq 1}$ is

$$\chi_j(t) = \begin{cases} \cos(j\pi t/2), & j \text{ odd}, \\ \sin(j\pi t/2), & j \text{ even}, \end{cases} \qquad T\chi_j = E_j\chi_j, \qquad E_j = (j\pi/2)^2.$$

Write $\chi = \chi_1 = \cos(\pi t/2)$. The transverse orthogonal projections are

$$P_T w = \chi_1 \int_J \overline{\chi_1(t)} w(t)\,dt, \qquad Q_T = I_{\mathfrak{h}} - P_T.$$

Their closed-channel space and the transverse restriction are

$$\mathfrak{h}_Q = \operatorname{Ran} Q_T = \{\chi_1\}^\perp = \overline{\operatorname{span}}\{\chi_j : j \geq 2\}, \qquad T_Q = T\big|_{D(T)\cap\mathfrak{h}_Q}.$$

The space $\mathfrak{h}_Q$ reduces T, so T_Q is selfadjoint in $\mathfrak{h}_Q$ and $T_Q \geq E_2$. All modes $j \geq 2$ are retained. Since $\pi^2 > 8$, the formula for λ_* gives $\lambda_* < E_2$.

On the half-strip and its unweighted auxiliary Hilbert space,

$$\mathcal{D} = (0, \infty) \times J, \qquad \mathfrak{H} = L^2(\mathcal{D}, ds\,dt) = L^2((0, \infty); \mathfrak{h}),$$

define the fibrewise extensions and their closed-channel range by

$$P_{\mathcal{D}} = I_{L^2(0,\infty)} \otimes P_T, \qquad Q_{\mathcal{D}} = I_{L^2(0,\infty)} \otimes Q_T, \qquad \mathfrak{H}_Q = \operatorname{Ran} Q_{\mathcal{D}} = L^2((0, \infty); \mathfrak{h}_Q).$$

These projections are orthogonal in the unweighted auxiliary spaces used for the divergence-form equation, not in the physical space $L^2(\mathcal{D}, f_\delta\,ds\,dt)$. The perturbed operator need not preserve this decomposition.

Put $\rho(s) = S + s$, with $S \geq 1$. Constants below may depend on the fixed k and λ_*, but are independent of δ and sufficiently large S. For $\mathcal{U} = (0, \infty)$ or $\mathcal{D}$ and $j \geq 0$ define

$$Y_j(\mathcal{U}) = \{F : \rho^j F \in L^2(\mathcal{U})\}, \qquad X_j(\mathcal{U}) = \{v : \rho^j \partial^\nu v \in L^2(\mathcal{U}),\ |\nu| \leq 2\},$$

with the norms

$$\|F\|_{Y_j(\mathcal{U})} = \left\|\rho^j F\right\|_{L^2(\mathcal{U})}, \qquad \|v\|_{X_j(\mathcal{U})} = \sum_{|\nu|\leq 2} \left\|\rho^j \partial^\nu v\right\|_{L^2(\mathcal{U})}.$$

All underlying norms use unweighted Lebesgue measure. On the half-line, ∂^ν denotes the longitudinal derivatives of orders $0, 1, 2$; on the half-strip it denotes all derivatives in s, t of total order at most two. Thus $X_0(\mathcal{U}) = H^2(\mathcal{U})$. Projection and boundary conditions are additional

conditions, not part of these definitions. For $\mathbb{R}^2$-valued functions we write $X_j((0,\infty);\mathbb{R}^2)$ and $Y_j((0,\infty);\mathbb{R}^2)$, using the same definitions with Euclidean norms. All constructions below preserve real-valued functions; we use their real subspaces when selecting the phase.

On $s \geq 0$ set

$$b_\delta(s) = \frac{16k}{\rho(s)}\left(1 + \frac{1}{2}\cos(2ks+\delta)\right), \qquad f_\delta(s,t) = \sqrt{1 + t^2 b_\delta(s)}. \tag{2}$$

For each fixed derivative order j,

$$|b_\delta^{(j)}(s)| \leq C_j \rho(s)^{-1}. \tag{3}$$

The eigenvalue equation for the stated form is exactly $\mathcal{L}_\delta \psi = 0$, where

$$\begin{aligned} \mathcal{L}_\delta &= -\partial_s(f_\delta^{-1}\partial_s) - \partial_t(f_\delta \partial_t) - \lambda_* f_\delta, \\ \mathcal{L}_{\text{free}} &= -\partial_s^2 - \partial_t^2 - \lambda_*, \qquad \mathcal{K}_\delta = \mathcal{L}_\delta - \mathcal{L}_{\text{free}}. \end{aligned} \tag{4}$$

On expanding derivatives, every coefficient of the order-two differential operator $\mathcal{K}_\delta$ is bounded by $C/\rho(s)$. They are combinations of $f_\delta^{-1} - 1$, $\partial_s f_\delta^{-1}$, $f_\delta - 1$, $\partial_t f_\delta$, and $\lambda_*(f_\delta - 1)$. Consequently

$$\|\mathcal{K}_\delta v\|_{Y_1(\mathcal{D})} \leq C \|v\|_{H^2(\mathcal{D})}, \tag{5}$$

$$\|\mathcal{K}_\delta v\|_{Y_1(\mathcal{D})} \leq \frac{C}{S}\|v\|_{X_1(\mathcal{D})}, \qquad \|\mathcal{K}_\delta v\|_{Y_2(\mathcal{D})} \leq C\|v\|_{X_1(\mathcal{D})}. \tag{6}$$

These statements do not require a longitudinal boundary condition on v. The coefficient maps in these operator norms are continuous and 2π-periodic in δ.

Let N be the longitudinal Neumann operator in $L^2(0,\infty)$:

$$N = -\frac{d^2}{ds^2}, \qquad D(N) = \{u \in H^2(0,\infty) : u'(0) = 0\}.$$

Lemma 1 (Free closed inverse)**.** *The operator*

$$A_0 = N \otimes I_{\mathfrak{h}_Q} + I_{L^2(0,\infty)} \otimes T_Q - \lambda_* I \quad \text{in } \mathfrak{H}_Q$$

has domain

$$D(A_0) = \{v \in H^2(\mathcal{D}) \cap \mathfrak{H}_Q : v(s,\pm 1) = 0,\ \partial_s v(0,t) = 0\},$$

with boundary conditions in the Sobolev trace sense. Its inverse satisfies

$$\left\|A_0^{-1}F\right\|_{H^2(\mathcal{D})} \leq C\|F\|_{L^2(\mathcal{D})}, \qquad \left\|A_0^{-1}F\right\|_{X_1(\mathcal{D})} \leq C\|F\|_{Y_1(\mathcal{D})},$$

for $F \in \mathfrak{H}_Q$, with the additional weighted condition in the second estimate, and with C independent of $S \geq 1$.

Proof. Expand $v(s,t) = \sum_{j\geq 2} v_j(s)\chi_j(t)$ in the transverse Dirichlet eigenbasis. The j-th component of $A_0 v$ is $-v_j'' + (E_j - \lambda_*)v_j$. Applying the continuous cosine transform in the longitudinal variable s, corresponding to Neumann data at zero, gives the multipliers

$$A_0 :\ \xi^2 + E_j - \lambda_*, \qquad A_0^{-1} :\ (\xi^2 + E_j - \lambda_*)^{-1}.$$

Here E_j comes from the discrete transverse expansion and ξ^2 from the continuous longitudinal transform. The spectral gap gives

$$\sup_{j\geq 2,\,\xi\geq 0} \frac{1 + \xi^2 + E_j}{\xi^2 + E_j - \lambda_*} < \infty.$$

This proves the unweighted estimate, including mixed derivatives, and identifies the domain as the stated closed-channel subspace of $H^2(\mathcal{D})$.

For the weighted estimate use $w(s) = S + \sqrt{1+s^2}$. Then $\rho \le w \le 2\rho$, $w'(0) = 0$, and $|w'| + |w''| \le C$. Choose bounded smooth truncations w_N tending pointwise to w, with $w_N \le w$, $w_N'(0) = 0$, and uniformly bounded first and second derivatives. Such truncations result by smoothly flattening $\sqrt{1+s^2}$ above level N before adding S. For $v = A_0^{-1}F$, multiplication by w_N preserves both boundary conditions and $\mathfrak{H}_Q$. Hence

$$A_0(w_N v) = w_N F - 2w_N' v_s - w_N'' v.$$

The unweighted estimate gives

$$\|w_N v\|_{H^2(\mathcal{D})} \le C\big(\|wF\|_{L^2(\mathcal{D})} + \|v\|_{H^2(\mathcal{D})}\big) \le C\,\|wF\|_{L^2(\mathcal{D})}\,.$$

The product rule bounds all $w_N \partial^\nu v$ for $|\nu| \le 2$ by the same right-hand side. Fatou's lemma proves the assertion. The weight chosen here preserves the Neumann condition at zero. □

Lemma 2 (Exact elimination of the closed modes)**.** *For sufficiently large S and every scalar $u \in H^2(0,\infty)$, there is a unique $z = z_\delta[u] \in X_1(\mathcal{D}) \cap \mathfrak{H}_Q$, with transverse Dirichlet conditions and $z_s(0,t) = 0$ such that*

$$Q_{\mathcal{D}}\mathcal{L}_\delta(u\chi + z) = 0 \quad \text{on } \mathcal{D}.$$

It satisfies

$$\begin{aligned} \|z_\delta[u]\|_{X_1(\mathcal{D})} &\le C\,\|u\|_{H^2(0,\infty)}\,, \\ \big\|\langle \chi, \mathcal{K}_\delta z_\delta[u]\rangle_{\mathfrak{h}}\big\|_{Y_2((0,\infty))} &\le C\,\|u\|_{H^2(0,\infty)}\,. \end{aligned} \tag{7}$$

The resulting linear maps are continuous and 2π-periodic in δ.

Proof. Let $R_0 = A_0^{-1}$. The equation for z is the fixed-point equation

$$z = -R_0 Q_{\mathcal{D}} \mathcal{K}_\delta(u\chi) - R_0 Q_{\mathcal{D}} \mathcal{K}_\delta z. \tag{8}$$

On the closed subspace of $X_1(\mathcal{D}) \cap \mathfrak{H}_Q$ carrying the stated boundary conditions, the second operator has norm at most C/S, by Lemma 1 and (6). The first term has $X_1(\mathcal{D})$ norm at most $C\,\|u\|_{H^2(0,\infty)}$ by (5). The Neumann series proves existence, uniqueness and the first bound. The second follows from the last estimate in (6). The lifted projection $Q_{\mathcal{D}}$ is bounded on the strip spaces used: it acts fibrewise in t, and the complementary transverse projection P_T has the fixed smooth range spanned by χ. Uniform convergence of the Neumann series proves parameter continuity.

No boundary condition on u is imposed here. This is an interior differential elimination with $z_s(0) = 0$. The full longitudinal boundary condition will be enforced later by $u'(0) = 0$. □

2 The exact scalar equation

Define the scalar functions

$$\begin{aligned} A(b) &= \int_J (1 + t^2 b)^{-1/2} \chi^2\, dt, \\ B(b) &= \int_J (1 + t^2 b)^{1/2} (\chi'^2 - \lambda_* \chi^2)\, dt, \\ \alpha(s) &= A(b_\delta(s)), \qquad \beta(s) = B(b_\delta(s)). \end{aligned} \tag{9}$$

Writing $\psi = u\chi + z_\delta[u]$, its remaining equation is precisely

$$-(\alpha u')' + \beta u + F_\delta u = 0, \qquad F_\delta u = \langle \chi, \mathcal{K}_\delta z_\delta[u]\rangle_{\mathfrak{h}}\,. \tag{10}$$

Here the inner product is taken in t, leaving a scalar function of s. Indeed, $P_{\mathcal{D}}\mathcal{L}_{\text{free}}z = 0$, and transverse integration by parts gives the coefficient β in (10).

The elementary identities

$$\int_J t^2\chi^2 = m, \qquad \int_J t^2\chi'^2 = E_1 m + 1$$

give, uniformly for small nonnegative b,

$$A(b) = 1 - \frac{m}{2}b + O(b^2), \qquad B(b) = -k^2 + \frac{1 - mk^2}{2}b + O(b^2). \tag{11}$$

The second identity follows by integrating $\int t^2\chi(-\chi'') = E_1 m$ once and using $\int 2t\chi\chi' = -1$.

Put $y = \sqrt{\alpha}\, u$. Direct differentiation gives

$$-(\alpha u')' + \beta u = \sqrt{\alpha}\,(-y'' - W_\delta y), \qquad W_\delta = -\frac{\beta}{\alpha} - \frac{\alpha''}{2\alpha} + \frac{(\alpha')^2}{4\alpha^2}.$$

Thus (10) is equivalent to

$$y'' + W_\delta y = \alpha^{-1/2} F_\delta(\alpha^{-1/2} y). \tag{12}$$

Taylor's formula for the smooth functions A, B, and (3), give

$$W_\delta = k^2 + (mk^2 - \tfrac{1}{2})b_\delta + \frac{m}{4}b_\delta'' + r_{1,\delta}, \qquad |r_{1,\delta}(s)| \le C\rho(s)^{-2}. \tag{13}$$

All remainders here are bounded by a constant times $b^2 + |bb''| + (b')^2$; no uncontrolled longitudinal derivative occurs. The choice of k gives $mk^2 = 1/2$. Differentiating (2) twice therefore yields

$$W_\delta = k^2 - \frac{4k}{\rho}\cos(2ks + \delta) + r_\delta, \qquad |r_\delta(s)| \le C\rho(s)^{-2}. \tag{14}$$

The complete equation, with every closed mode included, is

$$\mathfrak{L}_\delta y := y'' + \left(k^2 - \frac{4k}{\rho}\cos(2ks + \delta)\right) y = \mathcal{T}_\delta y, \tag{15}$$

where

$$\mathcal{T}_\delta y = \alpha^{-1/2} F_\delta(\alpha^{-1/2} y) - r_\delta y.$$

Since α stays bounded away from zero, its inverse square root and its first two derivatives are uniformly bounded. Lemma 2 and (14) show that

$$\mathcal{T}_\delta : H^2(0,\infty) \longrightarrow Y_2((0,\infty)), \qquad \|\mathcal{T}_\delta y\|_{Y_2((0,\infty))} \le C\,\|y\|_{H^2(0,\infty)}\,. \tag{16}$$

This is an exact bounded linear operator, generally nonlocal in s. It is real, continuous and 2π-periodic in δ in the indicated operator norm. Uniform bounds for the Taylor remainders and their δ derivatives give the same assertion for multiplication by r_δ.

3 A scalar right inverse with a small norm

Lemma 3. *For sufficiently large S there are real functions $h_\delta \in H^2(0,\infty)$ and real bounded linear maps $G_\delta : Y_2((0,\infty)) \to H^2(0,\infty)$ such that*

$$\mathfrak{L}_\delta h_\delta = 0, \qquad \mathfrak{L}_\delta G_\delta g = g, \qquad \|G_\delta\|_{Y_2((0,\infty))\to H^2(0,\infty)} \le C/S.$$

The family G_δ is continuous and 2π-periodic, while h_δ is continuous and satisfies $h_{\delta+2\pi} = -h_\delta$. For the vector $e_\delta \in \mathbb{R}^2$ and the scalar trace functional $\ell_\delta : H^2(0,\infty) \to \mathbb{R}$ defined by

$$e_\delta = \begin{pmatrix} \cos(3\pi/4 - \delta/2) \\ \sin(3\pi/4 - \delta/2) \end{pmatrix}, \qquad \ell_\delta(y) = e_\delta^{\mathsf{T}} \begin{pmatrix} y(0) \\ y'(0)/k \end{pmatrix},$$

the normalization is

$$\ell_\delta(h_\delta) = 1/S, \qquad \ell_\delta(G_\delta g) = 0. \tag{17}$$

Proof. Start with $\mathfrak{L}_\delta y = g$, with scalar $g \in Y_2((0,\infty))$, and put $v = (y, y'/k)^{\mathsf{T}} \in \mathbb{R}^2$. Let $R(s), E$ be the following real 2×2 matrices and $e_2 \in \mathbb{R}^2$:

$$R(s) = \begin{pmatrix} \cos ks & \sin ks \\ -\sin ks & \cos ks \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Writing $v = R(s)a$, with $a(s) \in \mathbb{R}^2$, converts the equation exactly to

$$a' = \frac{M_\delta + N_\delta(s)}{\rho} a + R(s)^{-1} e_2 g/k, \qquad M_\delta = \begin{pmatrix} \sin\delta & \cos\delta \\ \cos\delta & -\sin\delta \end{pmatrix}. \tag{18}$$

To verify this, $M_\delta + N_\delta(s)$ is the matrix $4\cos(2ks+\delta) R(s)^{-1} E R(s)$. Its mean over the period π/k is M_δ, so N_δ is bounded, periodic and has mean zero. Also $M_\delta^2 = I$ and $M_\delta e_\delta = -e_\delta$.

Define $Z_\delta(s) = \int_0^s N_\delta(r)\, dr$. It is bounded and periodic, with $Z_\delta(0) = 0$. For sufficiently large S, $H_\delta(s) = I + Z_\delta(s)/\rho(s)$ is uniformly invertible. These are real 2×2 matrices. Set $a = H_\delta x$, with $x(s) \in \mathbb{R}^2$. The exact transformed equation is

$$x' = \frac{M_\delta}{\rho} x + D_\delta(s) x + B_\delta(s) g, \tag{19}$$

where

$$D_\delta(s) = \rho^{-2} H_\delta^{-1}([M_\delta, Z_\delta] + N_\delta Z_\delta + Z_\delta),$$
$$B_\delta(s) = H_\delta^{-1} R(s)^{-1} e_2/k.$$

In particular $|D_\delta(s)| \le C\rho^{-2}$ and $|B_\delta(s)| \le C$. Here $D_\delta(s) \in \mathbb{R}^{2\times 2}$ and $B_\delta(s) \in \mathbb{R}^2$; both are continuous and 2π-periodic in δ.

The orthogonal projections $\Pi_\pm = (I_{\mathbb{R}^2} \pm M_\delta)/2$ act on $\mathbb{R}^2$, independently of the transverse and half-strip projections. For $F \in Y_2((0,\infty); \mathbb{R}^2)$ define

$$(\mathcal{I}_\delta F)(s) = \rho(s)^{-1} \int_0^s \rho(r) \Pi_- F(r)\, dr - \rho(s) \int_s^\infty \rho(r)^{-1} \Pi_+ F(r)\, dr. \tag{20}$$

This solves $x' = M_\delta x/\rho + F$ and has $\Pi_- x(0) = 0$. Moreover,

$$\|\mathcal{I}_\delta F\|_{L^2((0,\infty);\mathbb{R}^2)} \le \frac{C}{S} \|F\|_{Y_2((0,\infty);\mathbb{R}^2)}. \tag{21}$$

Indeed, writing $f = \rho^2 F \in L^2((0,\infty);\mathbb{R}^2)$, the stable kernel is $1_{r\le s}/(\rho(s)\rho(r))$, whose squared Hilbert–Schmidt norm is

$$\int_0^\infty \rho(r)^{-2} \int_r^\infty \rho(s)^{-2}\, ds\, dr = \frac{1}{2S^2}.$$

The unstable kernel is $1_{r\ge s}\rho(s)/\rho(r)^3$. Its row integrals are $1/(2\rho(s))$; its column integrals are $(\rho(r)^2 - S^2)/(2\rho(r)^3)$. Both are at most $1/(2S)$. The Schur estimate proves (21).

Multiplication by D_δ maps $L^2((0,\infty);\mathbb{R}^2)$ to $Y_2((0,\infty);\mathbb{R}^2)$ with uniformly bounded norm. Hence $I - \mathcal{I}_\delta D_\delta$ has a convergent Neumann inverse on $L^2((0,\infty);\mathbb{R}^2)$ for large S. Define

$$x_{h,\delta} = (I - \mathcal{I}_\delta D_\delta)^{-1}(\rho^{-1} e_\delta),$$
$$x_{g,\delta} = (I - \mathcal{I}_\delta D_\delta)^{-1}\mathcal{I}_\delta(B_\delta g).$$

These solve (19) with $g = 0$ and with the given g, respectively. They are in $H^1((0,\infty);\mathbb{R}^2)$: differentiating (20) in the distributional sense gives (19), whose right-hand side is in $L^2((0,\infty);\mathbb{R}^2)$. The integral formulas also give the trace data

$$\Pi_- x_{h,\delta}(0) = e_\delta/S, \qquad \Pi_- x_{g,\delta}(0) = 0.$$

Define h_δ and $G_\delta g$ as the first components of $RH_\delta x_{h,\delta}$ and $RH_\delta x_{g,\delta}$. Since all transformations were exact, the reconstructed vector satisfies

$$v_1' = kv_2, \qquad v_2' = -kv_1 + 4\rho^{-1}\cos(2ks + \delta)v_1 + g/k.$$

Thus the scalar functions belong to $H^2(0,\infty)$ and solve their asserted equations. In particular

$$\begin{aligned}\|G_\delta g\|_{H^2(0,\infty)} &\le C(\|x_{g,\delta}\|_{L^2((0,\infty);\mathbb{R}^2)} + \|g\|_{L^2(0,\infty)}) \\ &\le \frac{C}{S}\|g\|_{Y_2((0,\infty))},\end{aligned}$$

where $\|g\|_{L^2(0,\infty)} \le S^{-2}\|g\|_{Y_2((0,\infty))}$ was used. At zero, $R(0) = H_\delta(0) = I$, proving (17).

Every operator in these formulas is continuous and periodic in δ, while $e_{\delta+2\pi} = -e_\delta$. The claimed continuity and antiperiodicity follow, including in the $H^2(0,\infty)$ norm by the exact reconstructed equations. □

4 Phase selection and the weak eigenvalue identity

Take S larger than all the preceding fixed bounds. By (16) and Lemma 3, uniformly in δ we have

$$\|G_\delta \mathcal{T}_\delta\|_{H^2(0,\infty)\to H^2(0,\infty)} \le C/S < 1.$$

Thus

$$y_\delta = (I - G_\delta \mathcal{T}_\delta)^{-1} h_\delta \tag{22}$$

is well defined in $H^2(0,\infty)$ and satisfies (15). Its normalization remains $\ell_\delta(y_\delta) = 1/S$, so it is nonzero. It is real, continuous in δ, and $y_{\delta+2\pi} = -y_\delta$. Set

$$u_\delta = \alpha_\delta^{-1/2} y_\delta, \qquad z_\delta = z_\delta[u_\delta], \qquad \psi_\delta = u_\delta \chi + z_\delta.$$

These functions are real and antiperiodic in δ; u_δ is continuous in $H^2(0,\infty)$, and z_δ, ψ_δ are continuous in $H^2(\mathcal{D})$. The half-strip function ψ_δ solves $\mathcal{L}_\delta \psi_\delta = 0$ in $L^2(\mathcal{D})$ and has transverse Dirichlet boundary values. Its closed component satisfies $z_{\delta,s}(0) = 0$.

The real function

$$\mathcal{B}_S(\delta) = u_\delta'(0) = \alpha_\delta(0)^{-1/2}\left[y_\delta'(0) - \frac{\alpha_\delta'(0)}{2\alpha_\delta(0)} y_\delta(0)\right] \tag{23}$$

is continuous, since derivative trace is continuous on $H^2(0,\infty)$. It satisfies $\mathcal{B}_S(\delta+2\pi) = -\mathcal{B}_S(\delta)$. The intermediate value theorem gives $\delta_* \in [0, 2\pi]$ with $\mathcal{B}_S(\delta_*) = 0$. For definiteness, select the least zero in this compact interval. At this phase

$$\partial_s \psi_{\delta_*}(0,t) = 0. \tag{24}$$

Also $\psi_{\delta_*} \neq 0$: its $P_{\mathcal{D}}$ component is $\alpha_{\delta_*}^{-1/2} y_{\delta_*} \chi$ and $\ell_{\delta_*}(y_{\delta_*}) = 1/S$.

Extend b_{δ_*} and ψ_{δ_*} evenly to the full strip. The coefficient b is continuous and globally Lipschitz; a derivative jump at zero is permitted by the admissible class. Equation (24) ensures that the even extension of ψ belongs to $H^2(\mathbb{R} \times J)$ and that its longitudinal flux $f^{-1}\partial_s \psi$ has zero trace at the joining section. In particular $\psi \in H_0^1(\mathbb{R} \times J)$.

For any smooth compactly supported test function with zero transverse boundary values, integration by parts on both half-strips gives

$$\int_{\mathbb{R}\times J} \left(f^{-1} \psi_s \overline{\varphi_s} + f \psi_t \overline{\varphi_t} \right) ds\, dt = \lambda_* \int_{\mathbb{R}\times J} f \psi \overline{\varphi}\, ds\, dt. \tag{25}$$

The joining boundary terms vanish by (24). Since f and f^{-1} are bounded, both sides of (25) are continuous in the $H^1(\mathbb{R} \times J)$ norm of φ. Density proves this identity for every $\varphi \in H_0^1(\mathbb{R} \times J)$. The representation theorem for the defining form gives $\psi \in D(H_q)$ and $H_q \psi = \lambda_* \psi$.

Finally, the profile (1) is everywhere positive and

$$0 < q_{S,\delta_*}(s)^2 \leq \frac{24k}{S + |s|}.$$

Increasing S_0 to at least $12k$ proves $\|q\|_\infty \leq \sqrt{2}$ and uniform decay at both ends. Thus $q \in \mathcal{Q}_1$. The essential-spectrum theorem of [1, Theorem 3.1] gives $\sigma_{\mathrm{ess}}(H_q) = [E_1, \infty)$, so the eigenvalue is embedded. The cited source also supplies this essential-spectrum statement for the Lipschitz profile constructed here. For completeness, $f_{q_a}(s,t) = f_q(s/a, t/a)$, and

$$(U_a \psi)(s,t) = a^{-1} \psi(s/a, t/a)$$

is a unitary map between the two weighted L^2 spaces. Changing variables in the forms gives $h_{q_a}[U_a \psi] = a^{-2} h_q[\psi]$, hence $H_{q_a} U_a = a^{-2} U_a H_q$. This proves the assertion for every $a > 0$ and completes the proof of Theorem 1. □

Relation to the source and verification scope

For any ambient dimension $n+1 \geq 3$, choose

$$\Gamma(s) = (s, 0, \dots, 0), \qquad \theta(s) = \int_0^s q(r)\, dr, \qquad \Theta(s) = (\cos\theta(s), \sin\theta(s), 0, \dots, 0)$$

in the constant relatively parallel normal frame. The curve is unit speed and smooth, Θ is a Lipschitz unit vector, $|\Theta'| = q$ almost everywhere, and $k_{\mathrm{curve}} \cdot \Theta = 0$. The immersion is

$$\Phi(s,t) = (s, t\cos\theta(s), t\sin\theta(s), 0, \dots, 0).$$

Its metric is exactly $\mathrm{diag}(1 + t^2 q(s)^2, 1)$. It is injective: its first coordinate determines s, and at fixed s its unit transverse direction determines t. Thus it also gives a genuine embedded ribbon, although the repository permits an abstract immersed strip. Assumption 1 in [1] holds. Uniform decay gives its condition (3.1), and the amplitude bound is its condition (5.1). Theorem 3.1 and Corollary 5.3 give the essential spectrum; Theorem 5.1 supplies the stated Hardy inequality. The construction answers the point-spectrum part of Open Problem 5.6 negatively for the full class in AIM problem 1007.

The dated submission and literature audit accompanying this paper found no matching earlier resolution in the records checked on 3 November 2026. That bounded search does not establish publication priority. The existence argument above uses only elementary Fourier/Sobolev theory, convergent Neumann series, the intermediate value theorem, the form representation theorem, and the essential-spectrum theorem in [1].

References

[1] D. Krejčiřík and K. Zahradová, *Quantum strips in higher dimensions*, Operators and Matrices **14** (2020), 635–665. doi:10.7153/oam-2020-14-41. Published paper; arXiv:1906.02431.